Observation - Week 05

Topic: Full Stack Integration (MERN)

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Aim

To integrate a React frontend with an Express and MongoDB backend and build the complete ExpenseWise application.

Concepts Covered

- Project structure: client and server folders
- REST API consumed from React with fetch
- CORS and environment variables (dotenv)
- React state and effects (useState, useEffect)

Observations

- cors() must be enabled on the server because the client runs on a different port.
- Secrets such as the database URI stay in .env, which is excluded by .gitignore.
- The client reloads the list after every change so the UI always matches the database.
- The browser Network tab shows each API call with its status code and response.

Viva Questions

1 What is CORS?

2 What is the MERN stack?

3 Why use environment variables?

4 What does useEffect do?

Reference Files

- src/server/server.js
- src/server/package.json
- src/server/.env.example
- src/client/src/App.jsx
- src/client/src/main.jsx
- src/client/index.html
- src/client/package.json
- src/client/vite.config.js
- screenshots/frontend-ui-output.png
- screenshots/network-tab-api-call.png